From Surface Realism to Behavioral Validity: Rethinking Evaluation for Repurposed Interactive LLM Systems

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Abstract

Large language models (LLMs) are increasingly repurposed as interactive systems, including counselor agents, simulated clients, and patient agents. Yet their evaluation often remains closer to static, output-centered benchmarking than to the behavioral and human-facing claims attached to these systems. We examine this mismatch through a structured review of 52 LLM-based counseling, therapy, and behavioral simulation papers. Our analysis shows that evaluation is abundant but often local: papers frequently use automatic metrics, expert ratings, LLM judges, rubrics, and domain-aware criteria, yet less often test behavioral trajectories, intervention-sensitive dynamics, evaluator validity, or consequences for users. We also find that theory-related constructs commonly reach evaluation, but measurement remains uneven, often relying on paper-specific rubrics, simulated outcomes, or local quality judgments rather than standardized or claim-matched evidence. Building on this diagnosis, we propose behavior-grounded evaluation for repurposed interactive LLM systems. This framework links domain constructs to observable behavioral markers, tracks behavior over interaction, validates evaluator ecosystems, and matches evidence to the claims being made. We argue that NLP progress for repurposed interactive LLM systems should not be defined only by higher local scores, but by whether systems exhibit the intended behavior under meaningful interactional conditions with evidence appropriate to their human-facing use.

1 Introduction

Large language models (LLMs) are changing what it means to evaluate progress in NLP. For much of the field's history, progress has been organized around static benchmarks: fixed datasets, shared tasks, reference labels, and leaderboard metrics. This paradigm has been powerful because it makes model comparison tractable and cumulative. However, many current LLM systems are no longer used only as predictors over static inputs. General-purpose models are increasingly repurposed as interactive agents: counselors, simulated clients, patient agents, trainee partners, emotional-support systems, and role-play companions (Wang et al., 2024b, 2025b; Lee et al., 2024; Sungarda et al., 2025). In these settings, progress cannot be judged only by whether an output is fluent, preferred, or locally appropriate. It also depends on whether the system exhibits the behavior implied by its role, maintains that behavior across interaction, responds meaningfully to user actions, and supports the human-facing claims made about the system.

This creates a field-level evaluation mismatch. A simulated client may produce plausible utterances while drifting from its profile or responding inconsistently to counseling strategies (Yang et al., 2025b; Wang et al., 2026); a counseling agent may score highly on response quality while failing to maintain therapeutic stance or follow up on risk (Gabriel et al., 2024; Cai et al., 2026). The issue is therefore not that current work lacks evaluation. Rather, evaluation often remains local: it measures outputs, rubrics, and simulated interactions more readily than behavioral trajectories, intervention-sensitive dynamics, evaluator validity, or human-facing consequences.

LLM-based counseling, therapy, and role-play simulation provide a useful stress case for this mismatch. These systems are often motivated by goals such as counseling support, trainee practice, therapeutic usefulness, safety, or client simulation. They also rely on clinical, counseling, psychological, and behavioral constructs such as empathy, alliance, therapeutic stance, resistance, and risk handling (Li et al., 2024; Kim et al., 2025c). Many of these constructs are interactional: they unfold across turns, depend on user actions, and are interpreted relative to domain theory and use context. Recent reviews of mental-health LLMs and conversational agents highlight concerns around reliability, clinical applicability, safety, user experience, standardized evaluation, and longitudinal evidence (Hua et al., 2025; Bucher et al., 2025; Shao et al., 2025). However, existing reviews do not specifically diagnose how current NLP and HCI papers evaluate repurposed interactive LLM systems, or what kinds of evidence are used to support claims about behavior, training value, safety, and human-facing use.

We address this gap through a structured review of 52 LLM-based counseling, therapy, role-play, and behavioral simulation papers. Our goal is not to provide an exhaustive review of digital mental-health systems. Instead, we use this domain as an empirical case for a broader question: how should NLP evaluate progress when general-purpose models are reused as interactive systems? We analyze each paper along dimensions covering system design, evaluation authority, theory grounding, trajectory evidence, and human-facing outcomes.

Our analysis shows that evaluation in this area is abundant but unevenly aligned with system claims. Papers frequently use automatic metrics, expert ratings, LLM judges, rubrics, and domain-aware criteria (Xiao et al., 2024; Feng et al., 2025; Xu et al., 2025b). Yet these evaluations often provide evidence about local output acceptability rather than behavioral validity. We identify five recurring gaps: evaluation authority is broad but unevenly grounded; output quality is overused as a proxy for behavioral validity; theory reaches evaluation but measurement remains uneven; dynamic design outpaces trajectory evaluation; and human-facing motivations often outpace human-facing evidence.

Building on this diagnosis, we propose *behavior-grounded evaluation* for repurposed interactive LLM systems. Behavior-grounded evaluation asks whether a system exhibits the intended behavior over time, under meaningful interactional conditions, with evidence appropriate to the claim. It links domain constructs to observable behavioral markers, tests those markers across trajectories, probes intervention sensitivity, matches evidence to human-facing claims, validates evaluator ecosystems, and packages interaction protocols for reproducibility.

This paper makes three contributions. First, it provides a structured empirical diagnosis of evaluation practice in 52 LLM-based counseling, therapy, role-play, and behavioral simulation papers. Second, it identifies recurring gaps between current evaluation practices and the claims made about repurposed interactive LLM systems. Third, it articulates behavior-grounded evaluation as a claim-matched agenda for assessing interactive systems whose value depends on behavior over time, not only output quality. Together, these contributions argue that NLP progress in the LLM era should not be defined only by higher benchmark scores, but by whether evaluation can support the behavioral and human-facing claims attached to increasingly interactive systems.

2 Background: From Static Benchmarks to Repurposed Interactive LLM Systems

Prior reviews have established the rapid growth and difficulty of applying LLMs and conversational agents in mental-health contexts. Broad scoping reviews of mental-health LLMs highlight recurring concerns around reliability, clinical applicability, safety, and evaluation design (Hua et al., 2025). Integrative reviews of mental-health conversational agents further show that computer-science work often emphasizes model capability and response quality, whereas health-oriented work more often focuses on user outcomes and clinical relevance (Cho et al., 2023). Work on LLM-based medical-patient communication similarly emphasizes the gap between benchmark-centered model development and the interactional demands of clinical communication, where scenario alignment, communication quality, and human-centered evaluation become central (Shao et al., 2025). Together, these reviews motivate the importance of the domain, but they mainly characterize application areas, risks, capabilities, or broad evaluation challenges.

Our focus is different. We examine evaluation practice itself: who judges model behavior, what is measured, how theory is operationalized, whether dynamic behavior is evaluated over time, and whether evidence matches the human-facing claims made about these systems. This focus matters because LLM-based counseling and role-play systems are not only text generators. They are repurposed interactive systems: general-purpose models adapted through prompts, roles, memory, scaffolds, or interaction protocols to act as counselors, simulated clients, patient agents, or trainee partners. Their value depends not only on local response quality, but on whether they maintain coherent behavior, respond appropriately to user actions, and support claims about training, therapeutic usefulness, or simulation validity.

This concern is also visible in reviews focused on training, psychotherapy, and user experience. Reviews of conversational agents for mental-health training note challenges around standardized evaluation, authentic emotional responses, ethical safeguards, and clinical efficacy (Atapattu et al., 2025). Reviews of psychotherapy-oriented LLMs emphasize that assessment and intervention are dynamic and contextual, requiring systems that can adapt across stages of interaction and remain grounded in psychological theory (Na et al., 2025). Work on LLM mental-health applications also points to underexplored issues around design context, safety, user experience, and longitudinal outcomes (Bucher et al., 2025). These concerns make counseling and role-play simulation a useful stress case for a broader NLP evaluation problem.

We do not aim to provide an exhaustive review of digital mental-health systems. Instead, we use counseling and role-play simulation as a stress case for a broader NLP evaluation problem: how current NLP and HCI papers evaluate repurposed interactive LLM systems, and what kinds of evidence are needed when models are reused for behavioral, educational, therapeutic, or safety-relevant purposes. Existing reviews establish that evaluation is a recurring challenge; our review analyzes how evaluation is currently being done.

3 Method: Structured Review and Coding

We constructed and coded a corpus of LLM-based counseling and therapy papers, including closely related role-play and behavioral simulation work when it addressed client, patient, counselor, or psychologically grounded interaction. The review focused on papers that evaluated interactive generation, simulation behavior, counseling behavior, user experience, safety, or model performance.

3.1 Corpus construction

We searched ACL Anthology, ACM Digital Library, Google Scholar, and targeted HCI journals including IJHCS and IJHCI to cover NLP, HCI, and adjacent digital-health venues. Search queries combined terms for LLMs and generative agents with terms for counseling, psychotherapy, mental health, patient/client simulation, and evaluation. The initial search returned 900 records. After title/abstract screening, duplicate removal, and snowballing from eight highly relevant seed papers, we retained 112 candidate papers. Full-text relevance screening produced a final corpus of 52 primary papers. Figure 1 summarizes the corpus construction and screening process.

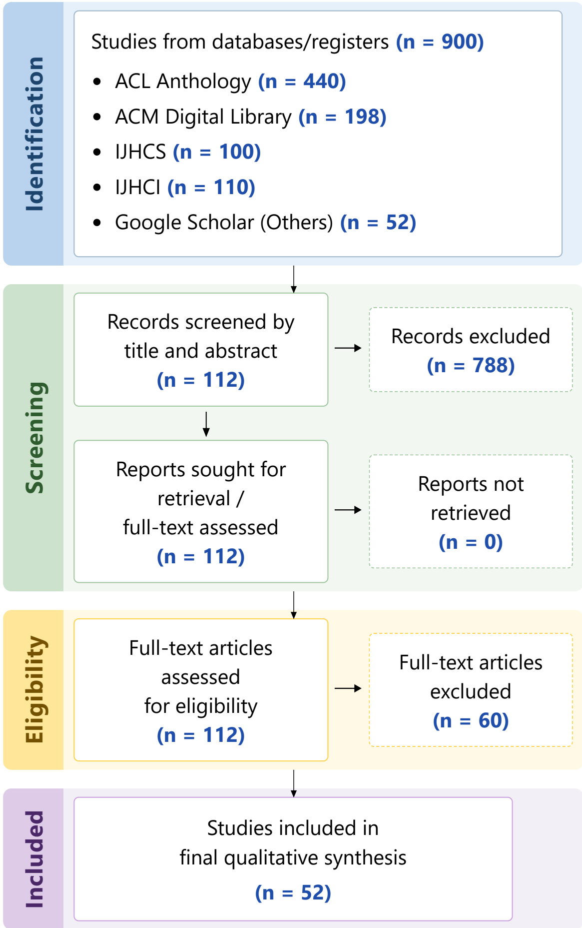

Figure 1: Corpus construction and screening process.

3.2 Inclusion criteria

We included papers that met three criteria: (1) they studied LLM-based or generative-agent systems; (2) they involved counseling, therapy, mental-health support, patient/client simulation, or behavioral/personality simulation; and (3) they reported some form of evaluation relevant to interaction, simulation behavior, counseling behavior, user experience, safety, or model performance. We excluded papers focused only on mental-health classification or detection, papers without interactive generation or simulation, papers outside the counseling simulation scope, and papers without accessible full text.

3.3 Coding scheme and validation

We iteratively developed a codebook with 38 fields covering simulation design, evaluation methods, human-centered targets, theory grounding, reliability/transparency, and robustness. Key fields included dynamic state, interaction level, longitudinal evaluation, evaluator type, LLM-judge use and validation, rubric use and reliability, theory grounding, theory operationalization, safety, realism, utility, fidelity, consistency, and human learning or outcome evidence.

Two student researchers coded the papers with ambiguous cases resolved through team discussion with the faculty lead. For fields central to our claims, we conducted additional audit passes to ensure that labels were supported by textual evidence.

4 Field Diagnosis: What Evaluation Measures and Misses

This section reports the main evaluation patterns in the 52-paper corpus and diagnoses what they leave under-tested. We organize the findings around five questions: who judges model behavior, what is measured, how theory enters evaluation, whether interactive behavior is evaluated over time, and whether evidence reaches the human-facing purposes used to motivate these systems. Across these dimensions, the central pattern is not an absence of evaluation. Rather, evaluation is abundant but often local: it measures outputs, rubrics, and simulated interactions more often than trajectories, consequences, or claim-specific evidence.

4.1 Evaluation authority is broad but unevenly grounded

The corpus shows a shift from single-metric evaluation toward evaluator triangulation. Automatic metrics appeared in 40 of 52 papers, human experts in 39, and LLM judges in 30; 21 papers combined all three. Examples include studies that combine expert or clinician ratings with LLM-based rubric scoring for counseling quality, safety, or therapeutic appropriateness (Qiu et al., 2025; Aghakhani et al., 2025; Cai et al., 2026), as well as studies that pair task-specific automatic metrics with human or LLM judgments for CBT, MI, or emotional-support behaviors (Xiao et al., 2024; Xu et al., 2025a; Gabriel et al., 2024). This is more informative than relying on automatic metrics alone, but it also requires clearer boundaries: automatic metrics, expert ratings, and LLM judges each support different kinds of evidence, and their scores should not be treated as interchangeable.

LLM-as-judge evaluation was common: 30 papers used LLM judges, and 26 reported some form of validation. This suggests that the field is not simply replacing human evaluation with unchecked LLM scores. The remaining issue is narrower but important: agreement with human raters can validate an LLM judge as a proxy scorer, but it does not by itself validate the rubric or the broader claim made from the score.

Rubric-based evaluation also creates a related issue. Forty-four papers used rubrics, but only 23 reported reliability or agreement (e.g., Cohen's kappa) for rubric-based judgments (Yang et al., 2025b; Feng et al., 2025). This matters because rubrics often define the central constructs being evaluated, such as empathy, safety, therapeutic appropriateness, or theory consistency. When reliability is not reported, it is difficult to know whether scores reflect stable criteria or evaluator-specific interpretations.

Thus, evaluation authority in this corpus is broad but unevenly grounded. The field increasingly triangulates among experts, LLM judges, and automatic metrics, but triangulation alone is not validation. The evidential force of these judgments depends on whether evaluators, rubrics, and validation procedures are aligned with the construct being measured and the claim being made.

4.2 Output quality is overused as a proxy for behavioral validity

The corpus shows substantial evaluation activity, but much of it remains close to the output surface. All 52 papers evaluated at least one local quality dimension, and 39 evaluated three or more. Utility was the most common target (46 papers), followed by fidelity (35), realism (30), emotional plausibility (30), consistency (22), and safety (22). These dimensions are necessary: unsafe, implausible, or task-unfaithful responses are clearly inadequate. The problem is that they mainly establish local acceptability.

Across papers, local acceptability is operationalized through measures such as empathy, coherence, guidance, authenticity, emotional expressiveness, active listening, relationship building, or therapy-technique use (Deshpande et al., 2025; Xie et al., 2025; Basar et al., 2025; Xu et al., 2025b). These measures can show that a response or short exchange looks appropriate. They provide weaker evidence about the system's behavior as an interactive agent, where validity depends on patterns that unfold across turns rather than on isolated judgments of local acceptability. In this sense, the field is not failing to evaluate; it is often asking output-level measures to stand in for behavioral evidence.

4.3 Theory reaches evaluation, but measurement remains uneven

Theory is not absent from this corpus; the sharper issue is how theory becomes measurement. Many papers incorporate clinical, counseling, psychological, or behavioral constructs into their framing, design, or evaluation. Forty of 52 papers showed strong theory grounding, and another 10 showed partial grounding. Theory-related constructs also appeared frequently in evaluation: 30 papers strongly operationalized such constructs, 16 did so partially, and 6 showed no clear theory-shaped evaluation.

The strongest cases made theory measurable through established psychological scales, therapy coding systems, theory-specific benchmarks, and construct-specific rubrics. Examples include therapeutic alliance and counseling-skill measures in CARE-Bench (Wang et al., 2026) and Client-CAST (Wang et al., 2024a), motivational interviewing criteria in KMI (Kim et al., 2025a) and CAMI (Yang et al., 2025a), and CBT-oriented ratings in CACTUS (Lee et al., 2024). Across the corpus, however, standardized instruments were less common than locally defined evaluation schemes: 9 papers used validated scales and 7 used established therapy or counseling coding systems, while 35 used theory-specific rubrics and 26 used theory-specific benchmark tasks. This suggests that the field often translates theory into evaluable dimensions, but not always through shared or standardized measurement instruments.

Even established instruments (e.g., PANAS, WAI) require caution in this setting. Measures of alliance, affect, empathy, symptoms, or counseling competence were originally designed for specific human respondents, trained observers, or clinical contexts. When such measures are answered by LLM-simulated clients or scored by LLM judges, the same score changes meaning: it may indicate simulated state consistency or agreement with local ratings, but not actual affective change, therapeutic alliance, user benefit, or safety. Thus, a recognized scale or rubric is not automatically strong evidence. It matters who produces the score and what claim the score is used to support.

The partial cases further show the unevenness of theory-shaped measurement. Sixteen papers relied on broader or paper-specific measures, including realism, authenticity, profile adherence, client resemblance, counseling quality, training usefulness, diagnostic correctness, or interaction quality (e.g., PsyCLIENT (Qiu et al., 2026), Roleplay-doh (Louie et al., 2024)). These measures are relevant to mental-health and counseling applications, but they are often defined differently across papers. As a result, it is difficult to compare results across studies or to know whether similarly named constructs are being measured in the same way. Thus, the gap is less the absence of theory-shaped evaluation than the unevenness of its measurement. Theory-related constructs are often evaluated, but not always in ways that are standardized, comparable, or valid for the specific evaluator, respondent, or claim being supported.

4.4 Dynamic design outpaces trajectory evaluation

For interactive counseling and role-play systems, behavioral validity depends on how behavior unfolds over time. A plausible multi-turn dialogue is not necessarily a valid behavioral trajectory: the system must maintain coherent profiles, states, and response patterns as the interaction develops. This matters for systems that claim memory, adaptation, client-state modeling, therapeutic progression, or sensitivity to counselor interventions.

Yet dynamic design outpaces dynamic evaluation. Twenty-seven of 52 papers studied extended dialogues, and 19 used some form of dynamic state, such as memory, evolving profiles, state transitions, session summaries, cognitive diagrams, or emotion trajectories. Only 7 papers reported longitudinal evaluation; among the 19 papers with dynamic state, 13 did not evaluate behavior longitudinally. Similarly, 39 papers involved intervention-sensitive evaluation or design, but only 7 of these included longitudinal evaluation. Thus, many papers examine whether a system can respond plausibly within an interaction, but fewer test whether those responses form stable, coherent, and theoretically meaningful behavioral trajectories.

A few papers point toward more trajectory-aware evaluation. EmoAgent measures pre–post changes in simulated psychological assessments (Qiu et al., 2025); the Interactive Narrative Therapist work evaluates therapeutic progression through innovative moments (Feng et al., 2025); and MusPsy uses multi-session counseling with working-alliance and affect measures across sessions (Wang et al., 2025a). These examples show that evaluation can track change across an interaction, even when the outcomes are simulated or proxy-based. However, they remain exceptions: dynamic behavior is often claimed or engineered, while evaluation still tends to reduce interaction to local judgments, aggregate scores, or single endpoint summaries.

4.5 Human-facing motivations often outpace human-facing evidence

The previous sections show that many papers evaluate outputs with domain-aware criteria. The remaining gap is whether such criteria are connected to the human-facing purposes used to motivate these systems, such as counseling support, trainee practice, therapeutic usefulness, or safety. User studies appeared in 19 of 52 papers, lay-user evaluation in 16, and explicit human learning or outcome measures in only 10. Across the corpus, 36 papers showed no clear evaluation beyond model outputs, benchmark performance, simulated dialogue quality, expert ratings, or LLM-judge scores.

This creates a claim–evidence gap. Utility was evaluated in 46 papers, but human learning or outcome evidence appeared in only 10. Safety was evaluated in 22 papers, but only a small number examined safety or deployment consequences. Similarly, 8 papers measured outcomes only in simulated clients or patients, such as simulated affect, alliance, symptom severity, or therapeutic progression. Such evidence is useful for testing synthetic interactions, but it should not be treated as evidence of real user benefit.

A few papers do connect evaluation to human-facing effects. Training-oriented studies measure counselor skill development (Louie et al., 2026; Tanana et al., 2019); perception-oriented studies measure perceived empathy or interaction experience (Schmidmaier et al., 2025b,a); and safety-oriented work evaluates high-risk mental-health dialogues (Cai et al., 2026). These examples remain exceptions. The core issue is not that every paper must evaluate all downstream effects, but that evidence should match the level of the claim: claims about training value, counseling support, therapeutic usefulness, or safety require more than local output quality or simulated outcomes.

5 Toward Behavior-Grounded Evaluation for Repurposed Interactive LLM Systems

The diagnosis above suggests that evaluation for repurposed interactive LLM systems should move from output-centered scoring to behavior-grounded evidence. We use *behavior-grounded evaluation* to refer to a layered evaluation logic: first define the domain construct being claimed, then identify observable behavioral markers, test whether those markers persist and change across interaction trajectories, probe whether they respond to meaningful interventions, and match the resulting evidence to the human-facing claim being made. This is not a call for every paper to run clinical trials or user studies. Rather, it requires authors to make explicit what level of claim their evidence supports. Output-quality claims, behavioral-simulation claims, training claims, safety claims, and deployment claims require different forms of evidence. The key shift is from asking whether an output looks good to asking whether a system exhibits the right behavior, under the right conditions, with evidence appropriate to the claim.

5.1 Construct-grounded evaluation

Evaluation should begin with the construct that a system claims to support or simulate. For repurposed interactive systems, broad labels such as "helpful", "realistic", or "therapeutic" are too underspecified to support strong claims. A counseling agent, for example, may need to demonstrate empathy, alliance-building, motivational interviewing adherence, CBT competence, or safety handling; a client simulator may need to represent avoidance, resistance, disclosure, emotional reactivity, or readiness to change. These constructs should be named, defined, and linked to the system claim before metrics are selected.

Construct-grounded evaluation therefore asks: what construct is the system expected to exhibit, and what would count as evidence for it? This does not require every study to use a validated clinical scale, but it does require making the evaluation target explicit. The key requirement is claim–construct alignment: the construct being measured should be the construct needed to support the paper's claim. Without such grounding, high realism or usefulness scores may obscure whether the system actually exhibits the intended behavior.

5.2 Behavioral marker evaluation

Constructs should be translated into behavioral markers: observable dialogue behaviors that indicate whether an abstract construct is being expressed. Empathy, for example, can be marked by validation, reflection, or perspective-taking; safety can be marked by risk assessment, boundary setting, or escalation behavior. This translation is the bridge between construct labels and dialogue evidence: it turns theory into inspectable behavior rather than leaving it as a label.

Behavioral marker evaluation makes evaluation more diagnostic. Instead of asking whether a dialogue is generally "good", "realistic", or "therapeutic", evaluators can ask whether expected behaviors occur, whether inappropriate behaviors are avoided, and whether the behavior matches the intended role. Without such markers, a dialogue can look plausible while still failing to exhibit the behavior the system is supposed to simulate or support.

5.3 Trajectory-based evaluation

Trajectory-based evaluation should define and track behavioral markers across an interaction. For systems that claim memory, adaptation, client-state modeling, or therapeutic progression, evaluation should specify how the relevant behavior is expected to persist, drift, or change as the interaction unfolds. A trajectory is therefore not simply a longer dialogue; it is an ordered behavioral path whose stability and development can be tested.

For a client simulator, evaluation should test whether the intended profile is preserved while relevant states change coherently across counselor actions, such as increased disclosure after empathic reflection or increased resistance after confrontation. For a counselor agent, evaluation should test whether the system maintains therapeutic stance, follows up on earlier disclosures, repairs rupture, handles risk consistently, or adapts intervention strategy over time. The unit of evaluation is not the isolated response, but the behavioral path that the system produces under interaction.

5.4 Intervention-sensitive evaluation

Intervention-sensitive evaluation should test behavioral contingencies, not merely response diversity. The evaluation should hold the client profile or scenario constant, vary the counselor action, and prespecify the expected behavioral effect of each intervention. For example, with the same avoidant client profile, empathic reflection might be expected to increase disclosure, confrontation might increase resistance, and cognitive reframing might shift interpretation of the problem. A behaviorally valid simulator should show these differentiated response patterns, rather than producing equally plausible continuations across all counselor actions.

This requires evaluating the effect of an intervention on subsequent markers, not only the immediate next response. For instance, after a risk disclosure, the system should show risk-assessment, boundary-setting, or escalation markers across the following turns, rather than generic reassurance. Thus, intervention-sensitive evaluation moves from asking whether a response sounds appropriate to asking whether meaningful actions produce theoretically expected behavioral consequences.

5.5 Claim-matched consequence evaluation

Claim-matched consequence evaluation links the strength of a claim to the level of evidence used to support it. Output-quality evidence can support claims about response generation or construct alignment; trajectory evidence can support claims about behavioral simulation; human-facing evidence is needed for claims about learning, support, or deployment readiness.

This distinction matters because evidence can be easily overextended. Simulated client outcomes can reveal synthetic interaction dynamics, but not real user benefit. Expert ratings can support clinical plausibility, but not necessarily trainee learning. LLM-judge scores can support scalable rubric scoring, but not user experience or safety in use. Behavior-grounded evaluation therefore requires making the boundary of evidence explicit: what the evaluation supports, what it does not support, and what stronger claims would require.

5.6 Validated evaluator ecosystems

Because repurposed interactive systems make different kinds of claims, no single evaluator is sufficient for all purposes. Automatic metrics can support scalable comparison, expert ratings can assess domain-specific plausibility, LLM judges can apply rubrics at scale, and users can report experience or perceived value. These sources of evidence should be treated as complementary rather than interchangeable.

A validated evaluator ecosystem specifies what each evaluator is trusted to judge. LLM judges should be calibrated against human or expert ratings for the specific rubric they apply, and human rubrics should report reliability when multiple raters are involved. More broadly, validation should operate at three levels: whether the evaluator can apply the rubric reliably, whether the rubric captures the intended construct, and whether the resulting score supports the claim being made. Triangulation is valuable, but triangulation alone is not validation; the evaluator must match the construct and claim.

5.7 Reproducible evaluation packages

For interactive systems, the evaluation object is not only the model output but the interaction protocol. Interactive evaluation is difficult to reproduce when papers disclose only final prompts, aggregate scores, or selected examples. For repurposed interactive LLM systems, reproducibility should include the evaluation protocol itself: role definitions, scenario seeds, prompts, state or memory updates, intervention scripts, dialogue sampling procedures, evaluator rubrics, LLM-judge prompts, human annotation guidelines, and failure cases.

Such packages would make evaluation more comparable across studies. They would allow researchers to test whether a system remains stable under the same client profile, responds consistently to the same intervention probes, or changes behavior under controlled perturbations. Reproducible evaluation packages are therefore not only a transparency measure; they are a prerequisite for cumulative progress in interactive NLP evaluation.

Together, these components turn the diagnosis from the previous section into a claim-matched evaluation workflow. They do not prescribe a single benchmark for all interactive LLM systems; instead, they specify how evidence should be constructed when systems are reused for behavioral, educational, therapeutic, or safety-relevant purposes.

6 Implications for NLP Progress

Our case study suggests that the question of progress in NLP changes when general-purpose models are repurposed as interactive systems. In static benchmark settings, progress can often be framed as better performance on a shared task. In reused human-facing systems, however, the meaning of progress depends on the claim being made: better response generation, valid behavioral simulation, training value, safety, therapeutic usefulness, or deployment readiness require different forms of evidence.

First, progress should be claim-relative. A higher local quality score is meaningful for claims about response generation, but it is weaker evidence for claims about behavioral simulation, training value, or human benefit. Our review shows that many papers evaluate outputs with domain-aware criteria, but evidence often remains local to model outputs, simulated interactions, or evaluator judgments. For repurposed interactive systems, progress should therefore be stated as a claim–evidence relation: what kind of improvement is claimed, and what kind of evidence would make that improvement meaningful?

Second, generalization should include behavioral transfer. For interactive LLM systems, generalization is not only performance on unseen examples or scenarios. It also concerns whether the system preserves the intended behavioral pattern across turns, users, interventions, and contexts. A client simulator that appears realistic in one dialogue but loses its profile under different counselor actions has not generalized behaviorally. A counseling agent that produces empathic responses but fails to maintain stance during rupture, resistance, or risk has not generalized interactionally.

Third, evaluation should become an ecosystem rather than a single score. Automatic metrics, expert ratings, LLM judges, rubrics, and user studies each provide different forms of evidence. The challenge is not simply to add more evaluators, but to define what each evaluator is trusted to judge and how those judgments are validated. Evaluator agreement may support local scoring reliability, but it does not automatically establish construct validity, trajectory validity, or human-facing benefit.

Overall, our findings suggest that benchmark performance is only one form of progress. For repurposed interactive LLM systems, progress also means producing behavior that is construct-grounded, stable across interaction, sensitive to meaningful interventions, and supported by evidence appropriate to the intended human use. This does not replace traditional evaluation; it extends it to settings where models are no longer only solving tasks, but participating in consequential interactions.

7 Conclusion

As LLMs are increasingly repurposed as interactive systems, NLP evaluation must account for more than local output quality. Using LLM-based counseling, therapy, and role-play simulation as a stress case, we reviewed 52 papers and found that evaluation is abundant but often local: studies frequently use automatic metrics, expert ratings, LLM judges, rubrics, and domain-aware criteria, yet less often test behavioral trajectories, intervention-sensitive dynamics, evaluator validity, or human-facing consequences.

The central issue is therefore not simply a lack of evaluation, but a mismatch between evidence and claim. Output-level or simulated evidence can show that responses look plausible, safe, empathic, or realistic, but it cannot by itself establish profile stability, therapeutic stance, trainee learning, user benefit, or deployment readiness. We argued for behavior-grounded evaluation as a claim-matched logic that links constructs to behavioral markers, tests them across trajectories and interventions, validates evaluator ecosystems, and makes evidence boundaries explicit.

For repurposed interactive LLM systems, progress should not be defined only by higher local scores. It should depend on whether systems exhibit the intended behavior, under meaningful interactional conditions, with evidence appropriate to the claims being made.

8 Limitations

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Category	Strong	Partial	Mentioned	Not Specified	Total
Theory Grounding	40 (76.9%)	10 (19.2%)	2 (3.8%)	—	52
Theory Operationalization	30 (57.7%)	16 (30.8%)	—	6 (11.5%)	52

Table 1: Statistics of Theory Grounding and Theory Operationalization in Evaluation

Category	Papers
Strong	(Xiao et al., 2024), (Gabriel et al., 2024), (Haydarov et al., 2025), (Qiu et al., 2025), (Aghakhani et al., 2025), (Wang et al., 2026), (Sungarda et al., 2025), (Chiu et al., 2024), (Li et al., 2024), (Wang et al., 2024a), (Fouda et al., 2026), (Cai et al., 2026), (Nguyen et al., 2025), (Zhang et al., 2025), (Kim et al., 2025a), (Jaiswal et al., 2024), (Xu et al., 2025a), (Schmidmaier et al., 2025b), (Schmidmaier et al., 2025a), (Ozgun et al., 2025), (Louie et al., 2026), (Wang et al., 2024b), (Lee et al., 2024), (Basar et al., 2024), (Yang et al., 2025b), (Yang et al., 2025a), (Xie et al., 2025), (Basar et al., 2025), (Feng et al., 2025), (Xu et al., 2025b), (Kim et al., 2025c), (Kim et al., 2025b), (Liu et al., 2025), (Wang et al., 2025c), (Samuel et al., 2025), (Chen et al., 2025), (Na, 2024), (Sharma et al., 2023), (Wang et al., 2025a), (Tanana et al., 2019)
Partial	(Wang et al., 2025b), (Liu et al., 2023), (Hu et al., 2025), (Qiu et al., 2026), (Deshpande et al., 2025), (Louie et al., 2024), (Kermani et al., 2025), (Kang et al., 2024), (Chen et al., 2023), (Schmidgall et al., 2025)
Mentioned	(Kuhail et al., 2025), (Berrezueta-Guzman et al., 2024)

Table 2: Papers Classified by Theory Grounding Level

Category	Papers
Strong	(Xiao et al., 2024), (Gabriel et al., 2024), (Qiu et al., 2025), (Aghakhani et al., 2025), (Wang et al., 2026), (Sungarda et al., 2025), (Chiu et al., 2024), (Li et al., 2024), (Wang et al., 2024a), (Cai et al., 2026), (Nguyen et al., 2025), (Zhang et al., 2025), (Kim et al., 2025a), (Xu et al., 2025a), (Schmidmaier et al., 2025b), (Schmidmaier et al., 2025a), (Louie et al., 2026), (Wang et al., 2024b), (Lee et al., 2024), (Yang et al., 2025a), (Feng et al., 2025), (Kim et al., 2025c), (Kim et al., 2025b), (Wang et al., 2025c), (Samuel et al., 2025), (Na, 2024), (Sharma et al., 2023), (Kang et al., 2024), (Wang et al., 2025a), (Tanana et al., 2019)
Partial	(Haydarov et al., 2025), (Liu et al., 2023), (Hu et al., 2025), (Fouda et al., 2026), (Qiu et al., 2026), (Jaiswal et al., 2024), (Ozgun et al., 2025), (Louie et al., 2024), (Basar et al., 2024), (Yang et al., 2025b), (Kermani et al., 2025), (Basar et al., 2025), (Liu et al., 2025), (Chen et al., 2025), (Chen et al., 2023), (Schmidgall et al., 2025)

Table 3: Papers Classified by Theory Operationalization in Evaluation

Evaluation Type	Example Papers
Validated scale *Standardized psychological or clinical scales*	(Xiao et al., 2024), (Qiu et al., 2025), (Li et al., 2024), (Wang et al., 2024a), (Schmidmaier et al., 2025b), (Schmidmaier et al., 2025a), (Yang et al., 2025a), (Chen et al., 2025), (Wang et al., 2025a)
Established therapy/counseling coding system *Therapy process or outcome coding*	(Xiao et al., 2024), (Zhang et al., 2025), (Lee et al., 2024), (Feng et al., 2025), (Kim et al., 2025c), (Kim et al., 2025b), (Tanana et al., 2019)
Theory-specific rubric *Custom rubrics based on theoretical constructs*	(Xiao et al., 2024), (Gabriel et al., 2024), (Qiu et al., 2025), (Aghakhani et al., 2025), (Wang et al., 2026), (Sungarda et al., 2025), (Liu et al., 2023), (Chiu et al., 2024), (Li et al., 2024), (Wang et al., 2024a), (Fouda et al., 2026), (Cai et al., 2026), (Qiu et al., 2026), (Nguyen et al., 2025), (Zhang et al., 2025), (Kim et al., 2025a), (Xu et al., 2025a), (Ozgun et al., 2025), (Louie et al., 2026), (Louie et al., 2024), (Wang et al., 2024b), (Lee et al., 2024), (Basar et al., 2024), (Yang et al., 2025b), (Yang et al., 2025a), (Basar et al., 2025), (Feng et al., 2025), (Kim et al., 2025c), (Kim et al., 2025b), (Samuel et al., 2025), (Chen et al., 2025), (Na, 2024), (Sharma et al., 2023), (Kang et al., 2024), (Wang et al., 2025a)
Custom expert rating *Expert judgments on custom dimensions*	(Xiao et al., 2024), (Gabriel et al., 2024), (Haydarov et al., 2025), (Wang et al., 2026), (Qiu et al., 2026), (Kim et al., 2025a), (Louie et al., 2026), (Louie et al., 2024), (Wang et al., 2024b), (Basar et al., 2024), (Yang et al., 2025b), (Basar et al., 2025), (Kim et al., 2025c), (Liu et al., 2025), (Na, 2024), (Sharma et al., 2023), (Kang et al., 2024), (Schmidgall et al., 2025), (Wang et al., 2025a)
Affective/emotional dynamics metric *Emotion recognition or affective computing metrics*	(Haydarov et al., 2025), (Schmidmaier et al., 2025a), (Wang et al., 2025c)
Clinical diagnostic benchmark *Clinical diagnosis or screening benchmarks*	(Qiu et al., 2025), (Hu et al., 2025), (Fouda et al., 2026), (Cai et al., 2026), (Ozgun et al., 2025), (Liu et al., 2025), (Chen et al., 2023), (Schmidgall et al., 2025)
Theory-specific benchmark/task *Task-specific performance benchmarks*	(Qiu et al., 2025), (Aghakhani et al., 2025), (Wang et al., 2026), (Sungarda et al., 2025), (Chiu et al., 2024), (Hu et al., 2025), (Fouda et al., 2026), (Nguyen et al., 2025), (Zhang et al., 2025), (Kim et al., 2025a), (Jaiswal et al., 2024), (Xu et al., 2025a), (Ozgun et al., 2025), (Wang et al., 2024b), (Lee et al., 2024), (Kermani et al., 2025), (Feng et al., 2025), (Kim et al., 2025b), (Liu et al., 2025), (Wang et al., 2025c), (Samuel et al., 2025), (Na, 2024), (Sharma et al., 2023), (Kang et al., 2024), (Chen et al., 2023), (Tanana et al., 2019)

Table 4: Evaluation Types Used in Included Studies

Evaluator Type	Count (%)	
Human Experts	34 (65%)	(Aghakhani et al., 2025), (Berrezueta-Guzman et al., 2024), (Cai et al., 2026), (Chen et al., 20
Lay Users	13 (25%)	
LLM Judges	30 (58%)	(Aghakhani et al., 2025), (Cai et al., 2026), (
Automatic Metrics	35 (67%)	(Aghakhani et al., 2025), (Basar et al., 2025), (Chen et al., 2023), (Chiu et al., 2024), (Deshpande et al., 202

Table 5: Distribution of Evaluator Types Across Studies ($N = 52$)

Has_Rubric	Reliability_Reported	Count	
Yes	Yes	13	
	No	31	(Basar et al., 2024), (Basar et al., 2025), (Berrezueta-Guzman et al., 2024), (Chen et al., 2023
No	Yes	2	
	No	3	
(unknown)	N/A	3	

Table 6: Scoring Rubric and Inter-Rater Reliability Reporting ($N = 52$)

Evaluation Dimension	Count (%)	
Safety	14 (27%)	
Realism	29 (56%)	
Consistency	20 (38%)	
Fidelity	33 (63%)	
Human Learning / Outcomes	10 (19%)	
Utility	44 (85%)	(Aghakhani et al., 2025), (Basar et al., 2024), (Basar et al., 2025), (Berrezueta-Guzman et al., 202
Emotional Plausibility	28 (54%)	

Table 7: Evaluation Dimension Coverage Across Studies ($N = 52$)

Interaction Level	Behavior Eval Depth	Count	
Single-turn	Static	8	(Berrezueta-
	Pattern-level	0	
	Dynamic	0	
	(unspecified)	4	
Short Multi-turn	Static	1	
	Pattern-level	0	
	Dynamic	0	
	(unspecified)	4	
Extended Dialogue	Static	16	(Chen et al., 2023), (Kang et al., 2024), (Kim et al., 2025a), (Kim et al., 2025c), (Lee e
	Pattern-level	7	(Che
	Dynamic	7	
	(unspecified)	1	
Longitudinal	Static	0	
	Pattern-level	0	
	Dynamic	2	

Table 8: Interaction Level by Behavioral Evaluation Depth ($N = 52$)

Human Learning	Lay Users	User Study	Count	
Yes	Yes	Yes	4	
		No	0	
	No	Yes	5	
		No	1	
No	Yes	Yes	5	
		No	4	
	No	Yes	3	
		No	30	(Aghakhani et al., 2025), (Cai et al., 2026), (Chiu et al., 2024), (Deshpande et al., 2

Table 9: Human Learning Outcomes by Lay User Involvement and User Study Design ($N = 52$)